

12 位高速 ADDA 直通实验

1、实验目的:

了解通过 12 位 ADS807 采样转成数据信号，再通过 12 位 DA 输出模拟信号，接示波器观察。

2、实验原理:

其中 ADS807/DA902 使用手册以提供 pdf 文档，供参考使用。本示例通过 AD 对外围提供的模拟信号进行采集，采集完的数据输送给 DA，转换成模拟信号输出。AD/DA 的工作时钟由 FPGA 提供，例:

```
4 module adda_direct( clk_50m, reset_n, ad_dat, da_dat, da_clk, ad_clk
5 );
6     input          clk_50m;
7     input          reset_n;
8     input [11:0]    ad_dat;
9     output          da_clk;
10    output          ad_clk;
11    output [11:0]    da_dat;
12    reg [11:0]       da_dat;
13    wire             clk_50m;
14    assign da_clk = ~clk_50m;
15    assign ad_clk = ~clk_50m;
16
17    always @ (posedge clk_50m or negedge reset_n)
18    begin
19        if(!reset_n)
20        begin
21            da_dat <= 12'b0;
22        end
23        else
24            da_dat <= ad_dat;
25    end
26 endmodule
27
```

3、硬件准备:

图 11-1ADDA 板介绍:

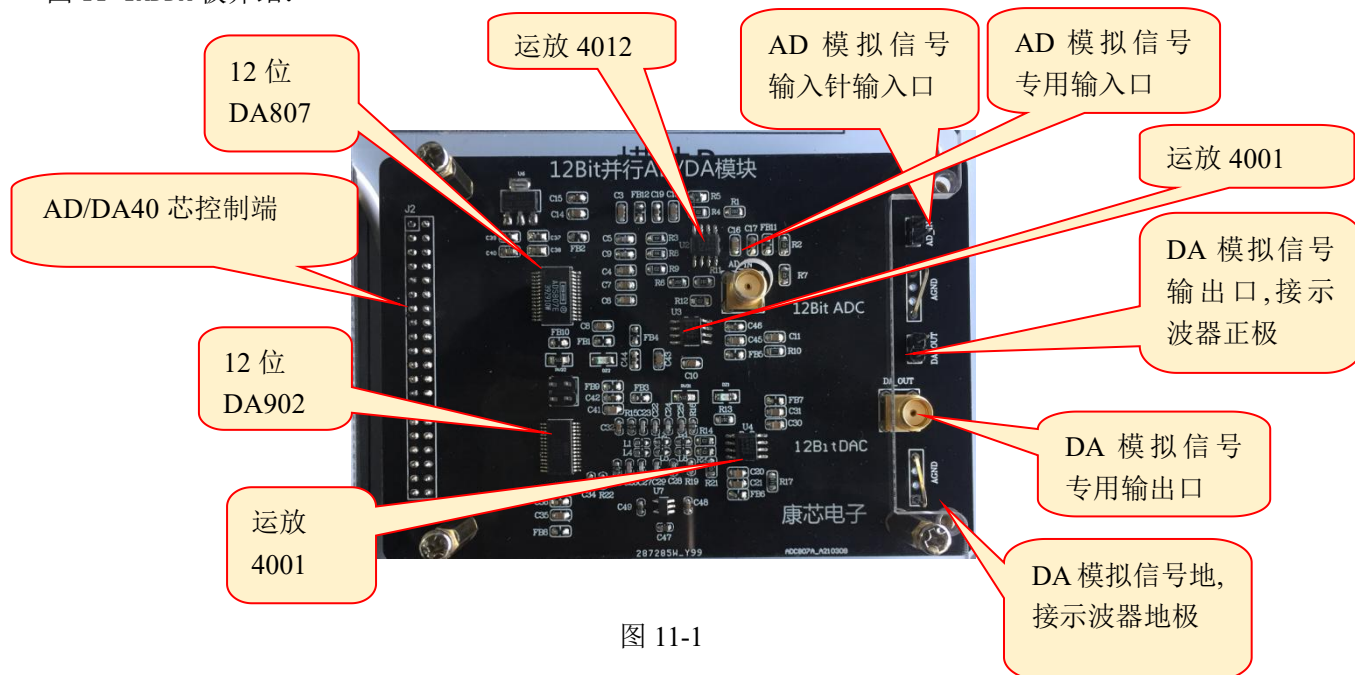


图 11-1

4、管脚约束:

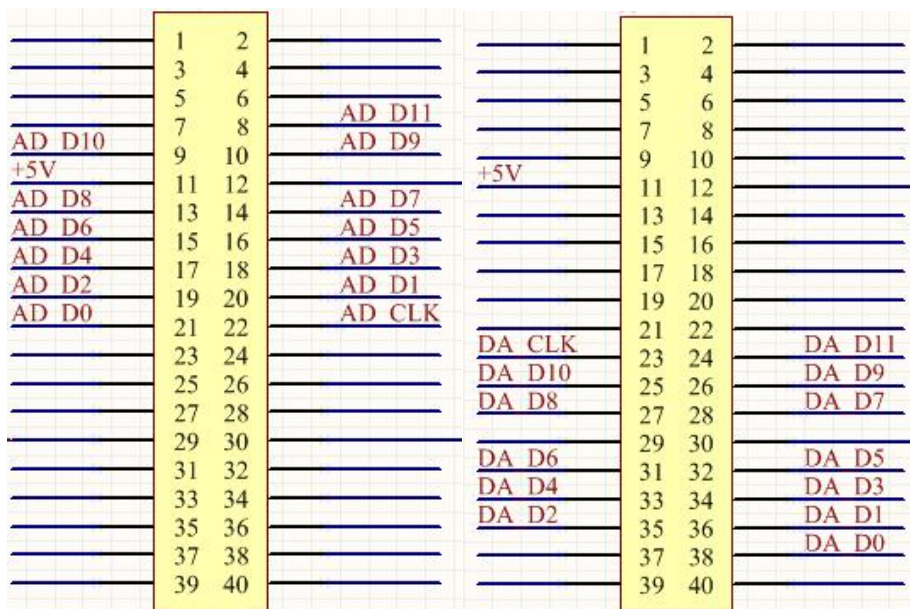


图 11-2

图 11-3

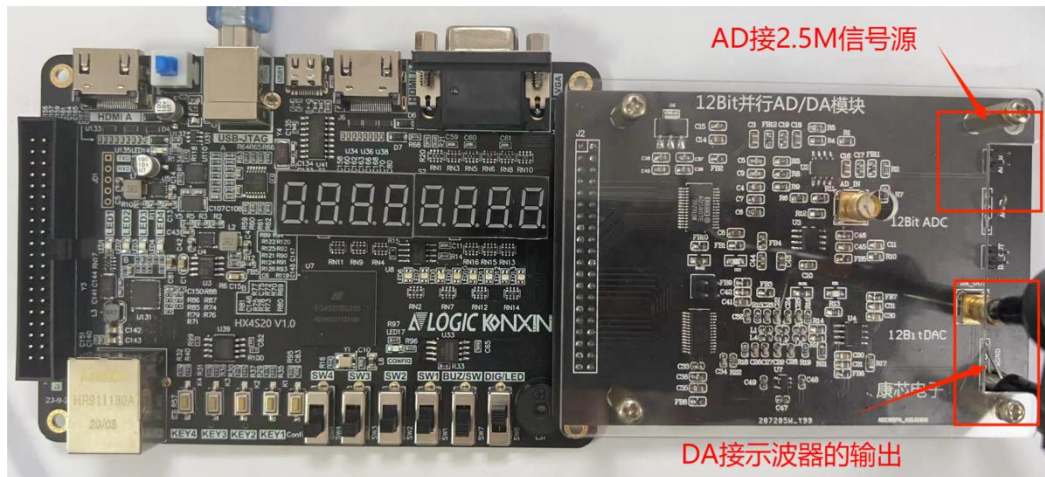
#IO 管脚约束

```
set_pin_assignment { ad_clk } { LOCATION = G14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;
PULLTYPE = NONE; }
set_pin_assignment { ad_dat[0] } { LOCATION = G16; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
```

```
set_pin_assignment { ad_dat[10] } { LOCATION = K12; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { ad_dat[11] } { LOCATION = J13; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { ad_dat[1] } { LOCATION = F15; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[2] } { LOCATION = F14; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[3] } { LOCATION = E15; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[4] } { LOCATION = D16; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[5] } { LOCATION = M10; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[6] } { LOCATION = L10; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[7] } { LOCATION = P11; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[8] } { LOCATION = M14; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { ad_dat[9] } { LOCATION = L14; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { clk_50m } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { clk_out } { LOCATION = P4; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;
PULLTYPE = NONE; }
set_pin_assignment { da_clk } { LOCATION = K15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;
PULLTYPE = NONE; }
set_pin_assignment { da_dat[0] } { LOCATION = R15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[10] } { LOCATION = H15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[11] } { LOCATION = J16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[1] } { LOCATION = P15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[2] } { LOCATION = P16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[3] } { LOCATION = N14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[4] } { LOCATION = N16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[5] } { LOCATION = L16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[6] } { LOCATION = M15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[7] } { LOCATION = M16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[8] } { LOCATION = K16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da_dat[9] } { LOCATION = H16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { reset_n } { LOCATION = A2; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
```

5、硬件连接:

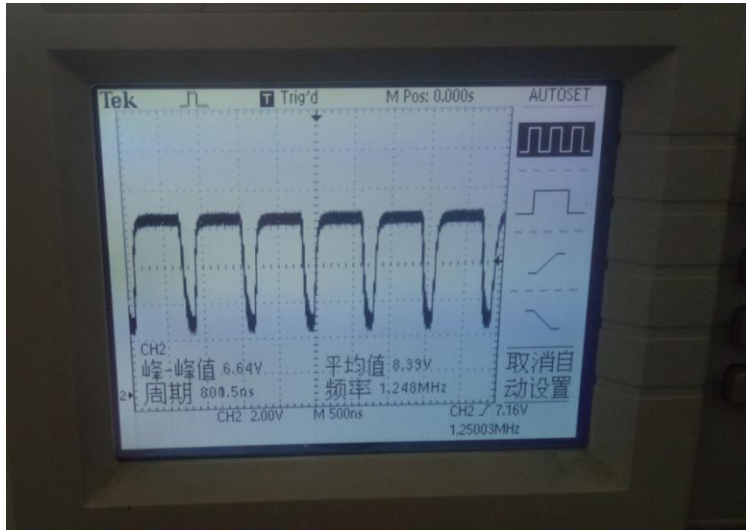
把 12 位高速 ADDA 模块插在开发板 7A75C 的 J1 口，下载程序后，通过信号源输入如：2.5MHz 晶振通过 AD_IN 接口，通过示波器接 DA_OUT 角查看转换波形。



观察示波器如 11-5，峰值不超过 500mv.

注意：操作注释事项

AD 模块的测试条件是输入峰峰值 500mV 的波形，不用偏置；



示波器观察波形，外围信号 2.5M